



6199 - Moving forward, get to the point: a forward model approach to the point cloud for accurate spectral extraction with JWST's IFUs

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

The goal of this proposal is to provide the JWST community with improved spectral extraction, PSF subtraction, and data modeling tools for the NIRSpec integral field units (IFU) and MIRI Medium Resolution Spectroscopy (MRS; also an IFU).

This program is motivated by the severe spatial undersampling of the JWST IFU modes, which is causing systematics in individual spatial pixels (spaxels) of the reconstructed spectral cubes (as high as 50% in single exposures) from the JWST calibration pipeline.

It was recently demonstrated that these systematics can be mitigated by fitting the data directly in detector images, also called the point cloud.

Building on recent literature work, we propose to develop tools for spectral extraction, PSF subtraction, and general forward modeling of the data in detector space.

Existing spectral extraction routines, based on interpolated spectral cubes and aperture photometry, only accurately work on isolated sources.

Aperture extraction is not adequate for more complex astrophysical scenes with overlapping signals because of the flux cross-contamination between neighboring sources. PSF fitting, or optimal extraction, is the preferred approach in these cases, but fitting data in the spectral cubes is hindered by the spaxel-to-spaxel systematics. Future IFU data analysis effort for JWST will benefit greatly from working directly in the point cloud, which is the goal of this program. This work will be especially relevant for any science case with IFU observations of blended sources, crowded fields, or high-contrast imaging, which include direct imaging of exoplanets, galactic center observations, gravitational lenses, to solar system objects.

OBSERVING DESCRIPTION